SA 220 AQUACULTURE

FACTORS TO CONSIDER WHEN SELECTING SPECIES FOR FISH FARMING

1. BIOLOGICAL CHARACTERISTICS OF AQUACULTURE SPECIES

- Rate of growth and production under culture conditions
- Faster growth rate, allows them to grow to marketable size in a shorter time, making it possible to have more frequent harvests
- size and age at first maturity is also an important consideration, as it will be preferable to have them reach marketable size before they attain first maturity so that most of the feed and energy are used for somatic growth.
- Early maturity good for breeders for hatchery operations but early maturity before the species reaches marketable size is a problem (tilapia species)

1. BIOLOGICAL CHARACTERISTICS OF AQUACULTURE SPECIES

- High fecundity can be an advantage
- Feeding behaviour of the species (herbivorous or omnivorous) species have been preferred
- Species that are hardy and can tolerate unfavourable conditions will have the advantage of better survival in relatively poor environmental conditions that may occur occasionally in culture situations
- Ability of the species to withstand high stocking densities in confinement is desired

1. BIOLOGICAL CHARACTERISTICS OF AQUACULTURE SPECIES

- Disease resistance
- Compatibility with other fish species in cultivation
- Conversion efficiency (feed-to-flesh)

2. ECONOMIC AND MARKET CONSIDERATIONS

- The availability of proven technologies of culture, backed by economic viability key in species selection for aquaculture
- Consumer acceptance and availability of markets for the species
- Improved markets developed, whether for domestic consumption or for export

3. INTRODUCTION OF EXOTIC SPECIES

- Some species have been selected for establishing sport and commercial fisheries, for ornamental purposes and for biological control
- Examples of species selected under this category include; tilapia species, cyprinids (common carp, Chinese carps), rainbow trout, walking catfish, Japanese and European oysters and fresh-water crayfish (Pacifastacus sp)
- Atlantic salmon, an exotic, has established itself so well in cage farming in Chile

3. criteria to be considered:

1. Fill a need, because of the absence of a similar desirable species in the locality of transplantation
2. Not compete with valuable native species to the extent of contributing to their decline;
3. Not cross with native species and produce undesirable hybrids
4. Not be accompanied by pests, parasites or diseases which might attack native species
5. Live and reproduce in equilibrium with its new environment

TAKE HOME QUESTION

1. List any five (5) key biological characteristics of aquaculture species that should be considered when selecting a species for farming

END

THANK YOU FOR YOUR ATTENTION